10TH CLASS CHEMISTRY — NEW SYLLABUS 2025-26 (PECTAA)

Chapter 22 — Hydrocarbons

Solved Quick Checks + Exercise | New Book (PECTAA 2025-26)

Exercise (Solved)

A. Multiple Choice Questions

1. Which of the following compounds is expected to give an addition reaction?

(a) $CH_3–CH_3$ (b) $CH_2=CH_2$
(c) CH_4 (d) $CH_3–CH_2–CH_3$

Answer: (b) — Ethene is an unsaturated hydrocarbon containing a carbon–carbon double bond, so a reactant can add across this bond; the other three are saturated alkanes.

2. Which of the following is not used as a fuel?

(a) LPG (b) CNG
(c) Diesel (d) Asphalt

Answer: (d) — Asphalt (bitumen) is a very heavy, non-volatile petroleum fraction used for making roads and waterproofing roofs, not as a fuel.

3. Indicate the type of the following reaction: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

(a) Substitution (b) Oxidation
(c) Reduction (d) Addition

Answer: (b) — Burning of methane in oxygen is oxidation (combustion), in which methane is oxidised to carbon dioxide and water.

4. Which species acts as a reducing agent in the reaction between ethene and $KMnO_4$?

(a) H_2O (b) $CH_2=CH_2$
(c) $KMnO_4$ (d) NaOH

Answer: (b) — Ethene is oxidised to ethane-1,2-diol and in doing so it reduces the permanganate ion, so ethene is the reducing agent.

5. The product which will be formed when propene reacts with bromine molecule.

(a) 1,2-Dibromopropane (b) 1,1-Dibromopropane
(c) 2,3-Dibromopropane (d) 1,3-Dibromopropane

Answer: (a) — Bromine adds across the double bond of propene, one bromine atom going to each of the two doubly bonded carbon atoms, giving $CH_3CHBrCH_2Br$ (1,2-dibromopropane).

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6. Select the name of the petroleum fraction which generates electricity in power plants.

(a) Bitumen (b) Lubricating oil
(c) Kerosene oil (d) Fuel oil

Answer: (d) — Fuel oil is used in ships and to run power plants for the generation of electricity.

7. The molecule eliminated in the following reaction is: CH_3–CH_2–Br + KOH → CH_2=CH_2 + KBr + H_2O

(a) Br_2 (b) H_2
(c) HBr (d) H_2O

Answer: (c) — Dehydrohalogenation removes a molecule of hydrogen bromide (HBr) from the halogenoalkane, which then forms a double bond.

8. Fractional distillation of petroleum is based on:

(a) Viscosity (b) Boiling point range
(c) Flammability (d) Melting point range

Answer: (b) — The different fractions of petroleum have different boiling point ranges, so they condense at different heights in the fractionating column.

B. Short Answer Questions

22.1 Give two examples of addition reactions.

(i) Addition of hydrogen (catalytic hydrogenation): when ethene is heated with hydrogen in the presence of nickel, ethane is formed.

$$CH_2=CH_2 + H_2 \rightarrow CH_3–CH_3 \quad \text{(Ni catalyst, about 150 °C)}$$

(ii) Addition of bromine: ethene reacts with bromine at room temperature to form 1,2-dibromoethane; the orange-brown colour of bromine water disappears.

$$CH_2=CH_2 + Br_2 \rightarrow CH_2Br–CH_2Br$$

22.2 Give the names of the compounds obtained from coal tar and petroleum.

From **coal tar** we obtain aromatic hydrocarbons such as benzene, toluene, naphthalene and phenol, along with dyes, drugs and perfumes which are prepared from them. From **petroleum** we obtain aliphatic hydrocarbons: the refinery gases (methane, ethane, propane and butane), petrol, naphtha, kerosene, diesel oil, fuel oil, lubricating oil and bitumen. Cracking of the heavier fractions gives smaller alkanes and alkenes such as ethene and propene, which are used to make plastics.

22.3 How ethene is oxidized by cold aqueous solution of potassium permanganate?

When ethene is shaken with cold, dilute, alkaline potassium permanganate solution, the purple colour of the permanganate disappears and ethene is oxidised to ethane-1,2-diol (ethylene glycol), which is a diol. In this reaction the permanganate ion is reduced to manganese dioxide, so a brown precipitate may also be seen. This reaction is used as a test for the presence of an alkene.

$$3CH_2{=}CH_2 + 2KMnO_4 + 4H_2O \rightarrow 3HOCH_2CH_2OH + 2MnO_2 + 2KOH$$

22.4 Differentiate between elimination and addition reactions.

Elimination reaction	Addition reaction
A small molecule such as water or hydrogen halide is removed from a molecule.	Atoms or groups are added across a double or triple bond of a molecule.
A saturated compound is converted into an unsaturated one, because a double or triple bond is formed.	An unsaturated compound is converted into a saturated one, because the multiple bond becomes a single bond.
Example: ethanol loses a molecule of water to give ethene (dehydration).	Example: ethene adds hydrogen to give ethane (hydrogenation).
It needs a reagent such as concentrated H_2SO_4 with heat, or alcoholic KOH.	It needs a reagent such as nickel catalyst, bromine water or steam with an acid catalyst.

22.5 Name the starting materials used to generate oxyacetylene flame.

The oxyacetylene flame is produced by burning acetylene (ethyne, C_2H_2) in pure oxygen. When acetylene is ignited in oxygen it burns with a very hot flame (about 3000 °C or more) which is used for welding and cutting metals.

$$2C_2H_2 + 5O_2 \rightarrow 4CO_2 + 2H_2O + \text{heat}$$

22.6 How can an alkene be identified?

An alkene can be identified by its reaction with bromine water. A little of the compound is shaken with bromine water in the dark, and if the orange-brown colour of bromine disappears quickly, the compound is an alkene, because bromine adds across its double bond. The same compound also decolourises cold, dilute, alkaline potassium permanganate solution and the violet colour of iodine. Saturated hydrocarbons do not give these reactions, so they are used to distinguish alkenes from alkanes.

C. Constructed Response Questions

22.1 What is the difference between the terms dehydration and dehydrohalogenation?

Dehydration	Dehydrohalogenation
A molecule of water is removed from the compound.	A molecule of hydrogen halide (HX) is removed from the compound.
It is carried out on an alcohol, using concentrated sulphuric acid as a dehydrating agent.	It is carried out on a halogenoalkane (alkyl halide), using potassium hydroxide dissolved in ethanol.
Example: ethanol gives ethene at about 180 °C: $CH_3CH_2OH \rightarrow CH_2{=}CH_2 + H_2O$.	Example: bromoethane gives ethene on heating with alcoholic KOH: $CH_3CH_2Br + KOH \rightarrow CH_2{=}CH_2 + KBr + H_2O$.
In both cases the molecule loses atoms from two neighbouring carbon atoms, so a double bond is formed; both are elimination reactions.	Both processes convert a saturated compound into an unsaturated one, so both are used to prepare alkenes.

22.2 You have got two test tubes A and B containing a saturated and an unsaturated compound (alkene). How will you identify the two?

Take a small amount of each compound in a test tube and add bromine water to both, then shake them and keep them away from strong light. The test tube in which the orange-brown colour of bromine water disappears quickly contains the **unsaturated compound (alkene)**, because bromine adds across its double bond. The test tube in which the colour of bromine water remains unchanged contains the **saturated compound**. This test can also be done with cold, dilute, alkaline potassium permanganate solution, whose purple colour is discharged only by the alkene.

22.3 Why is oxyacetylene flame so hot?

Acetylene (ethyne) is an unsaturated hydrocarbon which releases a very large amount of heat when it burns completely. In the oxyacetylene torch, acetylene is burnt in pure oxygen instead of air, so there is no nitrogen to carry away heat and dilute the flame. The complete combustion in pure oxygen produces a flame temperature of about 3000 °C, which is high enough to melt and join metals; therefore the flame is used for welding and cutting metals.

$$2C_2H_{2(g)} + 5O_{2(g)} \rightarrow 4CO_{2(g)} + 2H_2O(g) + \text{heat}$$

22.4 Do you expect ethyne to decolourize a cold solution of $KMnO_4$?

Yes. Ethyne is an unsaturated hydrocarbon because it contains a carbon–carbon triple bond, so it is oxidised by cold, dilute, alkaline potassium permanganate solution and the purple colour of $KMnO_4$ is discharged. This shows that the test with $KMnO_4$ is not limited to alkenes only; both alkenes and alkynes decolourise it, while alkanes do not.

22.5 How do chlorine and hydrogen chloride add to ethene?

Chlorine adds across the carbon–carbon double bond of ethene and forms 1,2-dichloroethane, in which one chlorine atom is attached to each of the two carbon atoms. Hydrogen chloride adds in a similar way: the hydrogen atom goes to one carbon and the chlorine atom to the other, giving chloroethane. In both reactions one of the two bonds of the double bond breaks and a single product is formed, so both are addition reactions.

$$CH_2=CH_2 + Cl_2 \rightarrow CH_2Cl–CH_2Cl \quad \text{(1,2-dichloroethane)}$$

$$CH_2=CH_2 + HCl \rightarrow CH_3–CH_2Cl \quad \text{(chloroethane)}$$

D. Descriptive Questions

22.1 Describe two methods of preparation for ethene.

(i) By dehydration of ethanol: ethanol is heated with concentrated sulphuric acid at about 180 °C. The acid acts as a dehydrating agent and removes a molecule of water from ethanol, so a double bond is formed and ethene is obtained. The ethene is collected over water.

$$CH_3CH_2OH \rightarrow CH_2=CH_2 + H_2O \quad \text{(conc. } H_2SO_4\text{, 180 °C)}$$

(ii) By dehydrohalogenation of a halogenoalkane: bromoethane (or chloroethane) is heated with potassium hydroxide dissolved in ethanol. Hydrogen and bromine are removed from the neighbouring carbon atoms, a double bond is formed and ethene is obtained, along with potassium bromide and water.

$$CH_3CH_2Br + KOH \rightarrow CH_2=CH_2 + KBr + H_2O \quad \text{(alcoholic KOH, heat)}$$

Ethene can also be prepared on a large scale by the cracking of the higher alkanes present in petroleum.

22.2 Describe the fractional distillation of petroleum.

Petroleum in the crude state is a complex mixture of many aliphatic hydrocarbons which do not burn easily, so it must be separated into useful fractions. This separation is called the refining of petroleum and it is carried out by fractional distillation.

First the crude oil is neutralised by washing it with an acidic or a basic solution as required. The neutralised oil is then heated in an electric furnace to above 400 °C, so most of it changes into vapour. These vapours are passed into a tall fractionating column which is hot at the bottom and cooler at the top, and which has a number of shelves with holes covered by bubble caps. As the vapours rise, the fraction with a high boiling point condenses first on the lower, hotter shelves, while the fractions with lower boiling points continue to rise and condense on the higher, cooler shelves. In this way each fraction is collected at a particular height of the column.

Fraction	Main use
Refinery gases	Used as gas for heating and cooking (LPG, CNG)

Fraction	Main use
Petrol (gasoline)	Fuel for cars and other motor vehicles
Naphtha	Raw material (chemical feedstock) for making chemicals
Kerosene (paraffin) oil	Jet fuel for aeroplanes
Diesel oil (gas oil)	Fuel for diesel engines
Fuel oil	Fuel for ships and for running power plants
Lubricating oil	Lubricants, waxes and polishes
Bitumen	Making roads and waterproofing roofs and pipes

Each fraction obtained is not a single pure compound but a mixture of hydrocarbons whose boiling points lie within a certain range.

22.3 Which properties of petroleum fractions change with their boiling points?

As we go from the lower fractions to the higher fractions of petroleum, the number of carbon and hydrogen atoms in the molecule increases, so the molecular mass increases and the intermolecular forces become stronger. Because of this the boiling point range and the viscosity increase, while the volatility and the flammability decrease.

Fractions with low boiling points	Fractions with high boiling points
They have shorter chains and smaller molecules.	They have longer chains and larger molecules.
Their molecular masses are small and the intermolecular forces are weak.	Their molecular masses are large and the intermolecular forces are strong.
They are more volatile and have low boiling point ranges.	They are less volatile and have high boiling point ranges.
They have low viscosity, so they flow easily, and they are highly inflammable.	They are more viscous and oily, and they burn with difficulty.
They are collected near the top of the fractionating column, for example refinery gases and petrol.	They are collected near the bottom of the column, for example lubricating oil and bitumen.

22.4 Explain with examples the cracking of alkanes.

Cracking is the process in which the higher alkanes of high molecular mass are broken into smaller, more useful alkanes and alkenes. For this purpose the alkane vapours are heated strongly to about 500 °C in the absence of oxygen and in the presence of zeolites, which act as catalysts. Cracking is important commercially because it changes less useful heavy fractions such as kerosene oil and diesel oil into petrol, alkenes and hydrogen gas, and the alkenes obtained are the starting materials for making plastics and many other chemicals.

Kerosene oil → Petrol + Alkenes + Hydrogen gas (catalyst, 500 °C)

$$C_{15}H_{32} \rightarrow 2C_2H_4 + C_3H_6 + C_8H_{18}$$

(ethene) (propene) (octane)

$$C_{10}H_{22} \rightarrow C_8H_{18} + C_2H_4$$ (decane → octane + ethene)

22.5 Give the uses of acetylene.

(i) Welding and cutting of metals: acetylene burns in oxygen with a very hot flame and is therefore used in the oxyacetylene torch for welding and cutting metals.

(ii) As a chemical feedstock: acetylene is an important starting material for the manufacture of many other organic chemicals and plastics.

(iii) Ripening of fruits: acetylene produced by the action of moisture on calcium carbide can accelerate the ripening of green fruits, but the use of calcium carbide for this purpose is prohibited in our country because it causes serious health problems such as dizziness, vomiting and skin ulcers. Ethene is the safe gas used for the controlled ripening of fruits.

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